LAPORAN TUGAS 7

PEMROGRAMAN SISI KLIEN

Identitas	Detail
Nama	Rizqy Akbar Dinul Putra
NIM	A11.2020.13139
Program Studi	Teknik Informatika

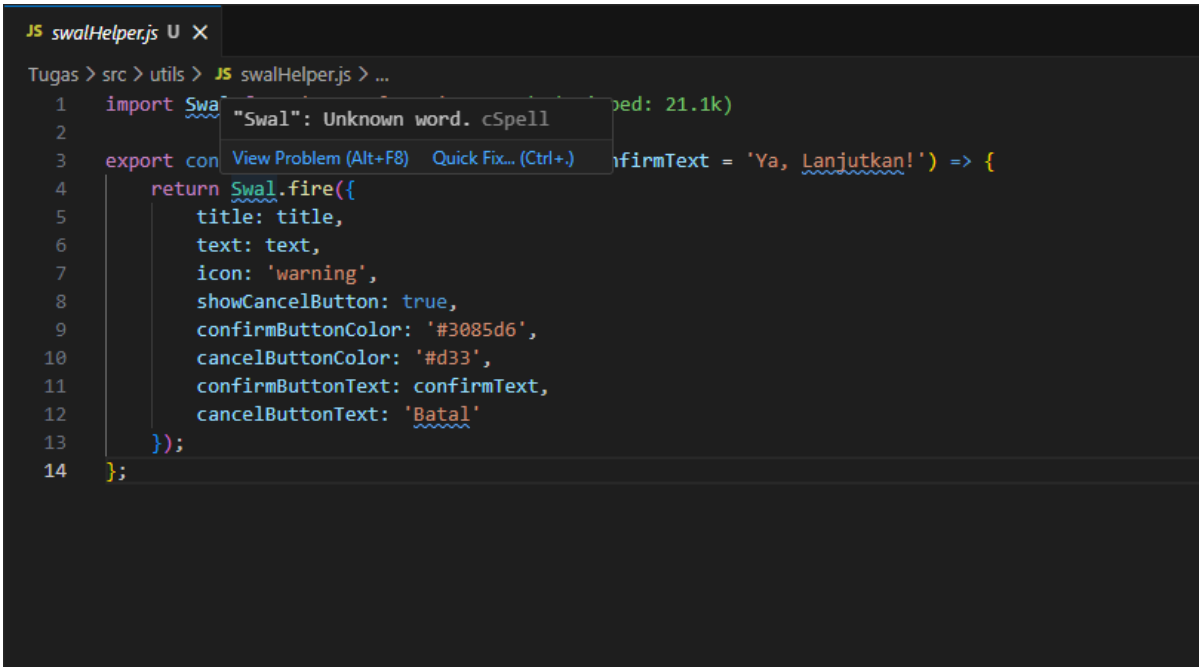
1. Tujuan Praktikum

- Memahami implementasi notifikasi modern pada React menggunakan library SweetAlert2 dan React Hot Toast.
- Membuat modul Helper khusus (`swalHelper` dan `toastHelper`) untuk mempermudah pemanggilan popup konfirmasi dan pesan umpan balik (feedback).
- Mengonfigurasi container Toaster global pada entry point React (`main.jsx`).
- Menggantikan dialog konfirmasi dan alert bawaan browser pada alur Login, Logout, Tambah, Edit, dan Hapus data Mahasiswa.

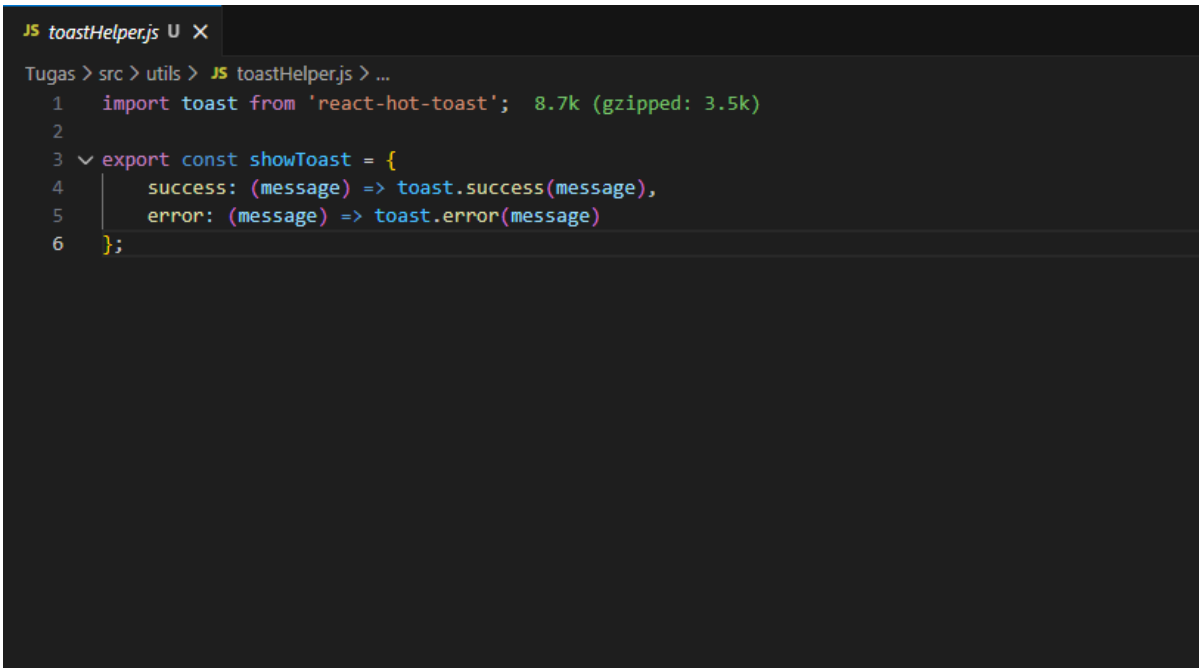
2. Source Code

Berikut adalah implementasi file pendukung dan komponen pada Tugas 7

A. Helper SweetAlert2



B. Helper Toast



C. Entry Point dan Setup Toast

main.jsx 1

Tugas > src > main.jsx > ...

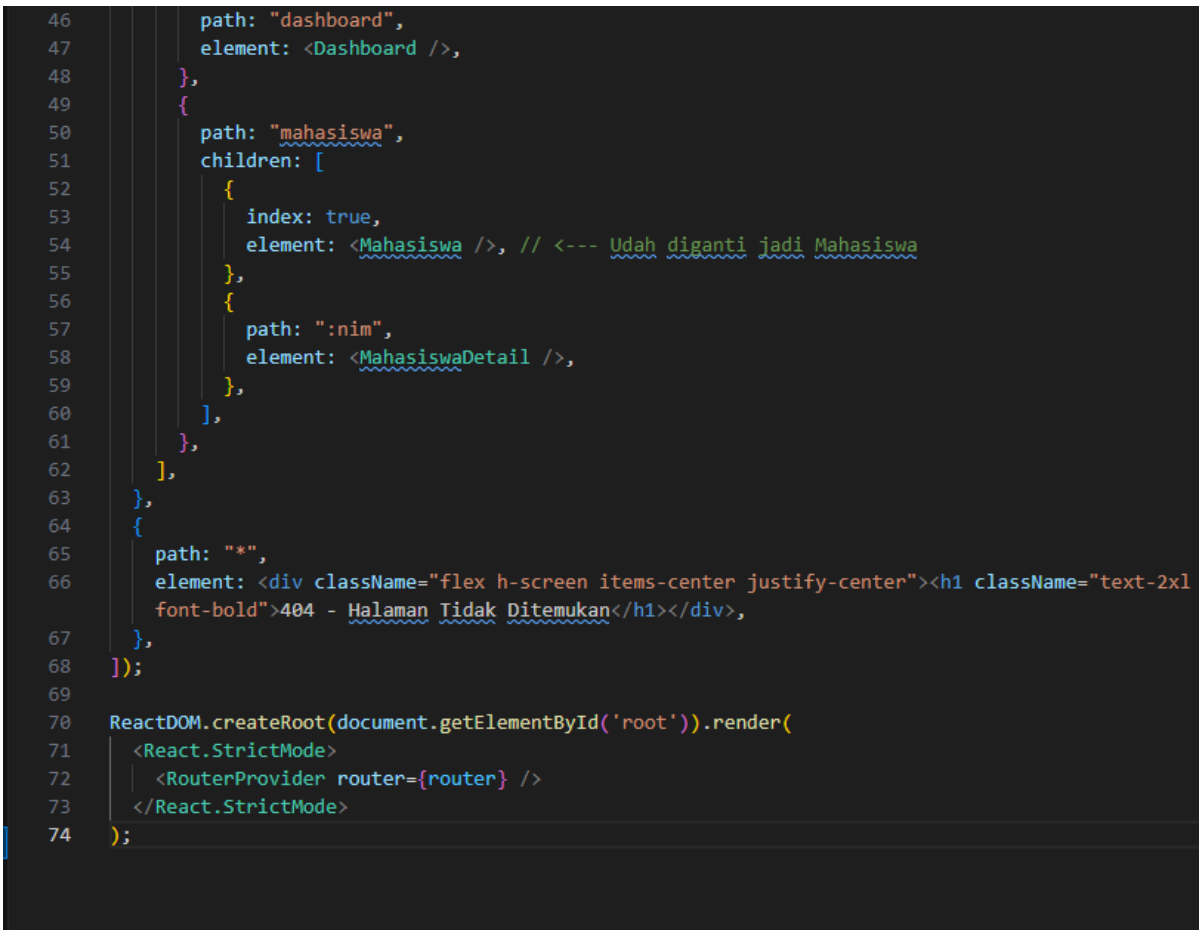
```jsx
import React from 'react';  8.2k (gzipped: 3.2k)
import ReactDOM from 'react-dom/client';  208.5k (gzipped: 65.3k)
import { createBrowserRouter, RouterProvider, Navigate } from 'react-router-dom';  60.6k (gzippec
import './index.css';

import AuthLayout from './layouts/AuthLayout';
import AdminLayout from './layouts/AdminLayout';

import Login from './pages/Login';
import Dashboard from './pages/Dashboard';
import Mahasiswa from './pages/Mahasiswa'; // Import Mahasiswa
import MahasiswaDetail from './pages/MahasiswaDetail';

const ProtectedRoute = ({ children }) => {
  const isLoggedIn = localStorage.getItem("user");
  if (!isLoggedIn) {
    return <Navigate to="/" />;
  }
  return children;
};

const router = createBrowserRouter([
  {
    path: "/",
    element: <AuthLayout />,
    children: [
      {
        index: true,
        element: <Login />,
      },
    ],
  },
  {
    path: "/admin",
    element: (
      <ProtectedRoute>
        <AdminLayout />
      </ProtectedRoute>
    ),
    children: [
      {
        index: true,
        element: <Navigate to="dashboard" />,
      },
      {
```



```
          path: "dashboard",
          element: <Dashboard />,
        },
        {
          path: "mahasiswa",
          children: [
            {
              index: true,
              element: <Mahasiswa />, // <--- Udah diganti jadi Mahasiswa
            },
            {
              path: ":nim",
              element: <MahasiswaDetail />,
            },
          ],
        },
      ],
    },
    {
      path: "*",
      element: <div className="flex h-screen items-center justify-center"><h1 className="text-2xl
      font-bold">404 - Halaman Tidak Ditemukan</h1></div>,
    },
  ]);

  ReactDOM.createRoot(document.getElementById('root')).render(
    <React.StrictMode>
      <RouterProvider router={router} />
    </React.StrictMode>
  );
```

D. Halaman Login

Login.jsx

Tugas > src > pages > Login.jsx > ...

```jsx
import Card from '../components/Card';
import Form from '../components/Form';
import Label from '../components/Label';
import Input from '../components/Input';
import Button from '../components/Button';

export default function Login() {

    // Data JSON Dummy sesuai materi dosen
    const dummyUser = {
        email: "admin@mail.com",
        password: "123"
    };

    const handleLogin = (e) => {
        e.preventDefault();
        const email = e.target.email.value; // Ambil nilai dari input id="email"
        const password = e.target.password.value; // Ambil nilai dari input id="password"

        // Cek apakah inputan
        if (email === dummyUs                    password) {
            // Kalau sukses, simpan 'user' ke localStorage
            localStorage.setItem("user", JSON.stringify(dummyUser));
            window.location.href = "/admin"; // Redirect cara dosen
        } else {
            alert('Email atau password salah bos!');
        }
    };

    return (
        <Card className="p-8 w-96">
            <h2 className="text-3xl font-bold text-center text-blue-600 mb-6">Login</h2>
            <Form onSubmit={handleLogin}>
                <div className="mb-4">
                    <Label htmlFor="email">Email</Label>
                    <Input
                        id="email"
                        name="email"
                        type="email"
                        placeholder="admin@mail.com"
                        required
                    />
                </div>
                <div className="mb-4">
                    <Label htmlFor="password">Password</Label>
                    <Input
                        id="password"
                        name="password"
                        type="password"
                        placeholder="123"
                        required
                    />
                </div>
                <Button type="submit" className="w-full mt-4">Login</Button>
            </Form>
        </Card>
    );
}
```

"simpan": Unknown word. cSpell

View Problem (Alt+F8) Quick Fix... (Ctrl+.)

E. Komponen Sidebar dan Logout

Sidebar.jsx M

Tugas > src > components > Sidebar.jsx > Sidebar

```jsx
import { NavLink, useNavigate } from 'react-router-dom';  10.9k (gzipped: 4.3k)
import { confirmAlert } from '../utils/swalHelper';

export default function Sidebar() {
    const navigate = useNavigate();

    const handleLogout = () => {
        confirmAlert('Logout', 'Yakin mau keluar dari aplikasi?', 'Ya, Logout!').then((result)
        => {
            if (result.isConfirmed) {
                localStorage.removeItem('user');
                navigate('/');
            }
        });
    };

    return (
        <div className="w-64 bg-blue-800 text-white flex flex-col min-h-screen">
            <div className="p-6">
                <h2 className="text-2xl font-bold">Admin</h2>
            </div>
            <nav className="flex-1 mt-4">
                <ul className="space-y-2">
                    <li>
                        <NavLink
                            to="/admin/dashboard"
                            className={({ isActive }) =>
                                `flex items-center px-6 py-3 hover:bg-blue-700
                                transition-colors ${isActive ? 'bg-blue-700 border-l-4
                                border-white' : ''}`
                            }
                        >
                            <span className="mr-3">📊</span> Dashboard
                        </NavLink>
                    </li>
                    <li>
                        <NavLink
                            to="/admin/mahasiswa"
                            className={({ isActive }) =>
                                `flex items-center px-6 py-3 hover:bg-blue-700
                                transition-colors ${isActive ? 'bg-blue-700 border-l-4
                                border-white' : ''}`
                            }
                        >
                            <span className="mr-3">🎓</span> Mahasiswa
                        </NavLink>
                    </li>
                </ul>
            </nav>

            <div className="p-4 border-t border-blue-700">
                <button onClick={handleLogout} className="w-full bg-red-600 hover:bg-red-700
                text-white py-2 px-4 rounded font-bold transition-colors">
                    🚪 Logout
                </button>
            </div>
        </div>
    );
}
```



F. Halaman Mahasiswa

Mahasiswa.jsx M

Tugas > src > pages > Mahasiswa.jsx > Mahasiswa

```jsx
import { useState } from 'react';  5k (gzipped: 2.1k)
import Card from '../components/Card';
import Button from '../components/Button';
import MahasiswaTable from '../components/MahasiswaTable';
import MahasiswaModal from '../components/MahasiswaModal';
import { confirmAlert } from '../utils/swalHelper';
import { showToast } from '../utils/toastHelper';

export default function Mahasiswa() {
    const [mahasiswa, setMahasiswa] = useState([
        { nim: 'A11.2020.13139', nama: 'Rizqy Akbar Dinul Putra', status: true }
    ]);
    const [selectedMahasiswa, setSelectedMahasiswa] = useState(null);
    const [isModalOpen, setModalOpen] = useState(false);

    const storeMahasiswa = (data) => {
        const isNimExists = mahasiswa.find(m => m.nim === data.nim);
        if (isNimExists) {
            showToast.error("NIM sudah terdaftar! Gunakan NIM lain.");
            return;
        }
        setMahasiswa([...mahasiswa, data]);
        showToast.success("Data mahasiswa berhasil ditambahkan!");
    };

    const updateMahasiswa = (data) => {
        const isNimExists = mahasiswa.find(m => m.nim === data.nim);
        if (isNimExists && isNimExists.nim !== selectedMahasiswa.nim) {
            showToast.error("NIM sudah terdaftar pada mahasiswa lain!");
            return;
        }

        confirmAlert('Simpan Perubahan?', 'Apakah Anda yakin ingin memperbarui data mahasiswa
        ini?', 'Ya, Simpan!').then((result) => {
            if (result.isConfirmed) {
                setMahasiswa(mahasiswa.map(m => m.nim === selectedMahasiswa.nim ? data : m));
                showToast.success("Data berhasil diupdate!");
                setModalOpen(false);
                setSelectedMahasiswa(null);
            }
        });
    };

    const deleteMahasiswa = (nim) => {
        setMahasiswa(mahasiswa.filter(m => m.nim !== nim));
    };
```

Mahasiswa.jsx M

Tugas > src > pages > Mahasiswa.jsx > Mahasiswa

```jsx
export default function Mahasiswa() {

    const openAddModal = () => {
        setModalOpen(true);
        setSelectedMahasiswa(null);
    };

    const openEditModal = (mhs) => {
        setModalOpen(true);
        setSelectedMahasiswa(mhs);
    };

    const handleSubmit = (data) => {
        if (selectedMahasiswa) {
            updateMahasiswa(data);
        } else {
            storeMahasiswa(data);
            setModalOpen(false);
            setSelectedMahasiswa(null);
        }
    };

    const handleDelete = (nim) => {
        confirmAlert('Hapus Data?', 'Yakin ingin menghapus data mahasiswa ini secara permanen?
        ', 'Ya, Hapus!').then((result) => {
            if (result.isConfirmed) {
                deleteMahasiswa(nim);
                showToast.success("Data mahasiswa berhasil dihapus!");
            } else {
                showToast.error("Penghapusan dibatalkan.");
            }
        });
    };

    return (
        <>
            <Card className="p-6">
                <div className="flex justify-between items-center mb-4">
                    <h3 className="text-lg font-bold text-gray-700">Daftar Mahasiswa</h3>
                    <Button onClick={openAddModal}>+ Tambah Data</Button>
                </div>

                <MahasiswaTable
                    mahasiswa={mahasiswa}
                    openEditModal={openEditModal}
                    onDelete={handleDelete}
                />
            </Card>

            <MahasiswaModal
                isModalOpen={isModalOpen}
                onClose={() => { setModalOpen(false); setSelectedMahasiswa(null); }}
                onSubmit={handleSubmit}
                selectedMahasiswa={selectedMahasiswa}
            />
        </>
    );
}
```

G. Komponen Modal Mahasiswa

MahasiswaModal.jsx 1, M

Tugas > src > components > MahasiswaModal.jsx > MahasiswaModal

```jsx
import { useState, useEffect } from 'react';  5.1k (gzipped: 2.1k)
import Button from './Button';
import Input from './Input';
import Label from './Label';
import { showToast } from '../utils/toastHelper';

export default function MahasiswaModal({ isModalOpen, onClose, onSubmit, selectedMahasiswa }) {
    const [form, setForm] = useState({ nim: '', nama: '', status: true });

    useEffect(() => {
        if (selectedMahasiswa) {
            setForm(selectedMahasiswa);
        } else {
            setForm({ nim: '', nama: '', status: true });
        }
    }, [selectedMahasiswa, isModalOpen]);

    const handleChange = (e) => {
        const { name, value, type, checked } = e.target;
        setForm(prev => ({
            ...prev,
            [name]: type === 'checkbox' ? checked : value
        }));
    };

    const handleSubmit = (e) => {
        e.preventDefault();

        if (!form.nim.trim() || !form.nama.trim()) {
            showToast.error("NIM dan Nama tidak boleh kosong!");
            return;
        }

        onSubmit(form);
    };

    if (!isModalOpen) return null;

    return (
        <div className="fixed inset-0 bg-black bg-opacity-50 flex items-center justify-center
        p-4 z-50">
            <div className="bg-white rounded-lg p-6 w-full max-w-md shadow-lg">
                <h3 className="text-xl font-bold mb-4">{selectedMahasiswa ? 'Edit Mahasiswa' :
                'Tambah Mahasiswa'}</h3>
                <form onSubmit={handleSubmit}>
                    <div className="mb-4">
```

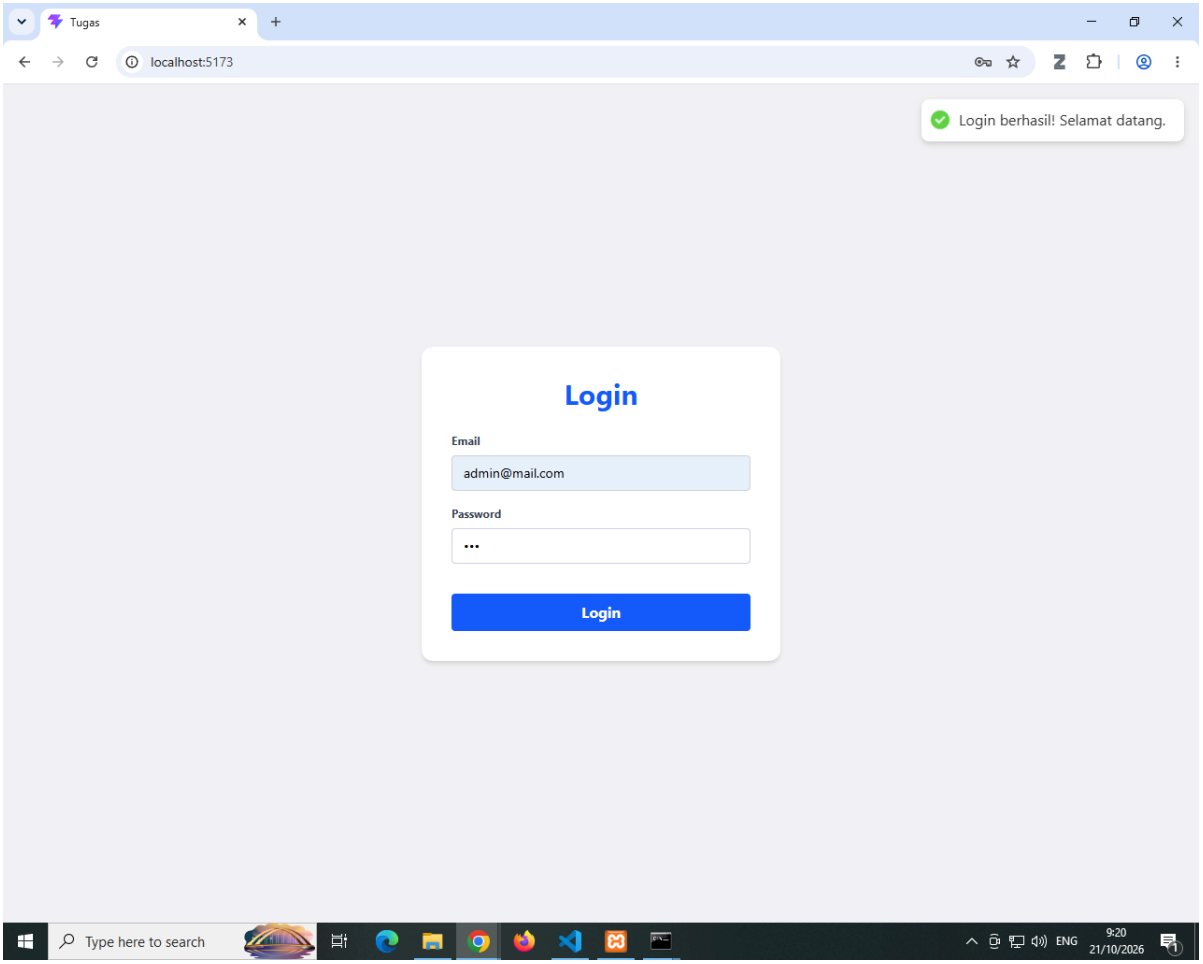
MahasiswaModal.jsx 1, M

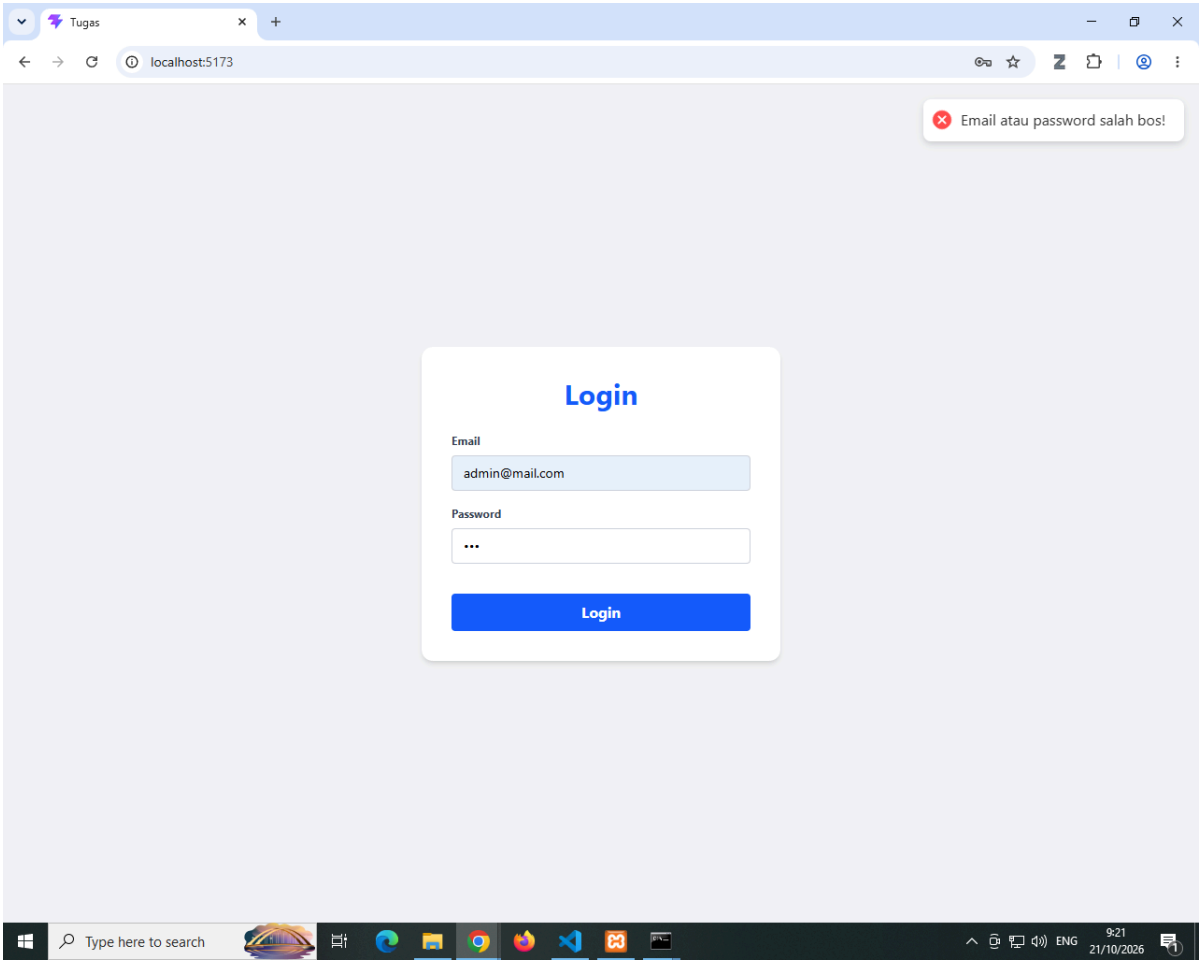
Tugas > src > components > MahasiswaModal.jsx > MahasiswaModal

```jsx
export default function MahasiswaModal({ isModalOpen, onClose, onSubmit, selectedMahasiswa }) {
                        <Label htmlFor="nim">NIM</Label>
                        <Input
                            type="text"
                            id="nim"
                            name="nim"
                            value={form.nim}
                            onChange={handleChange}
                            placeholder="Masukkan NIM"
                        />
                    </div>
                    <div className="mb-4">
                        <Label htmlFor="nama">Nama</Label>
                        <Input
                            type="text"
                            id="nama"
                            name="nama"
                            value={form.nama}
                            onChange={handleChange}
                            placeholder="Masukkan Nama Lengkap"
                        />
                    </div>
                    <div className="mb-6">
                        <Label htmlFor="status" className="flex items-center cursor-pointer">
                            <input
                                type="checkbox"
                                id="status"
                                name="status"
                                checked={form.status}
                                onChange={handleChange}
                                className="mr-2 h-4 w-4 text-blue-600 focus:ring-blue-500
                                border-gray-300 rounded"
                            />
                            <span className="text-gray-700">Status Aktif</span>
                        </Label>
                    </div>
                    <div className="flex justify-end space-x-2">
                        <Button type="button" onClick={onClose} className="bg-gray-500
                        hover:bg-gray-600">Batal</Button>
                        <Button type="submit" className="bg-blue-600 hover:bg-blue-700">Simp
                        Data</Button>
                    </div>
                </form>
            </div>
        </div>
    );
```

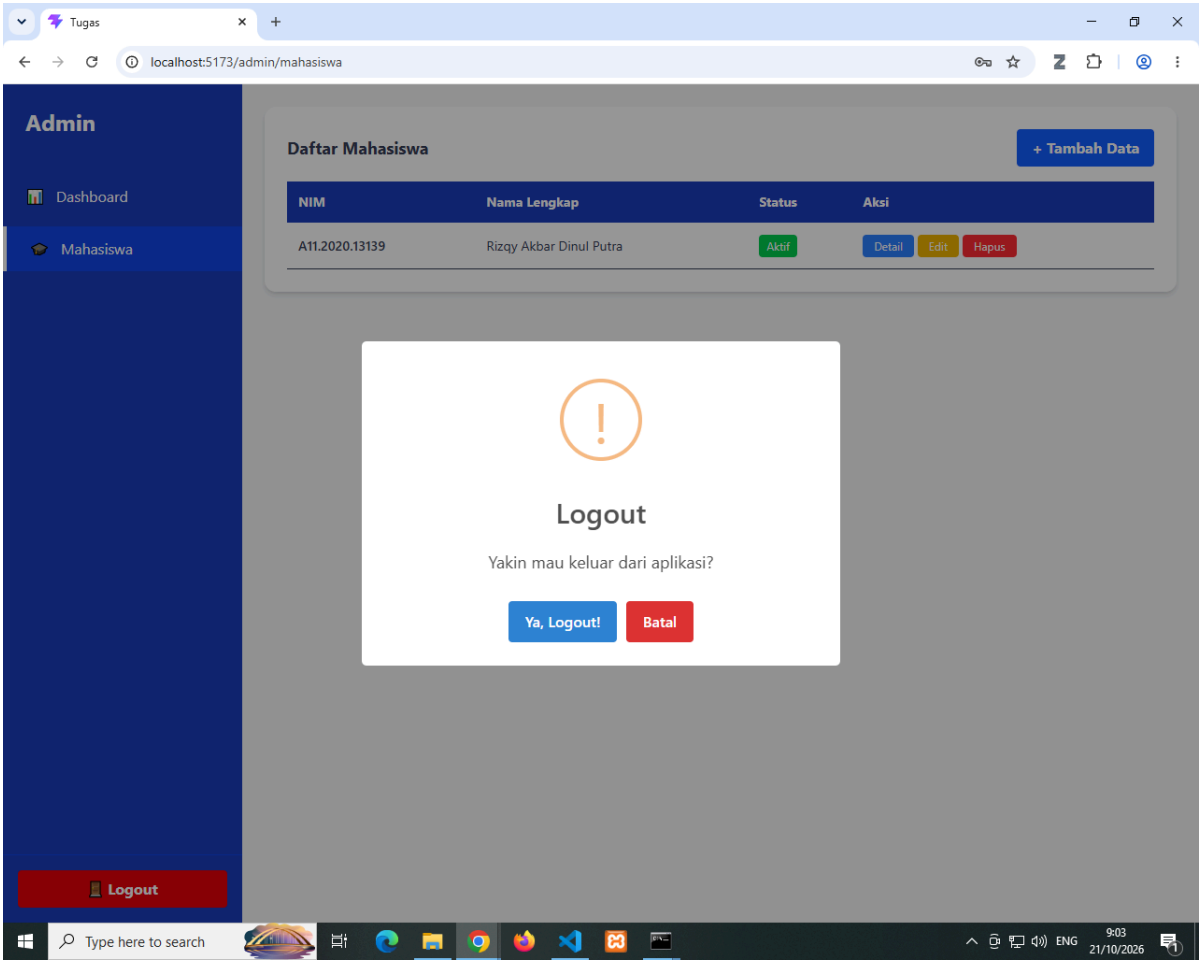
3. Hasil Tampilan dan Pengujian

A. Toast Notifikasi Login (Berhasil atau Gagal)

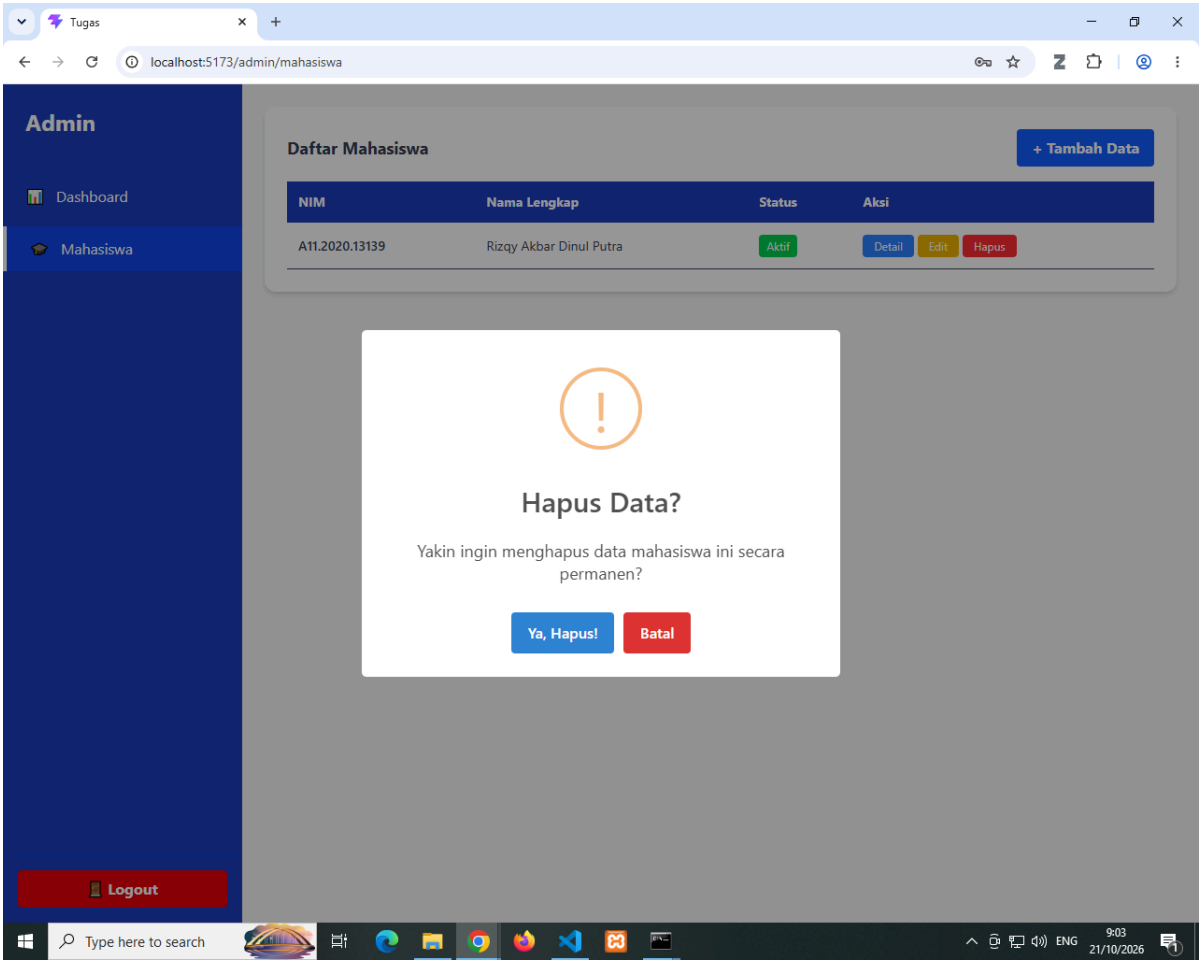

Tugas
localhost:5173
Login berhasil! Selamat datang.
Login
Email
admin@mail.com
Password
...
Login
Type here to search
ENG
9:20
21/10/2026



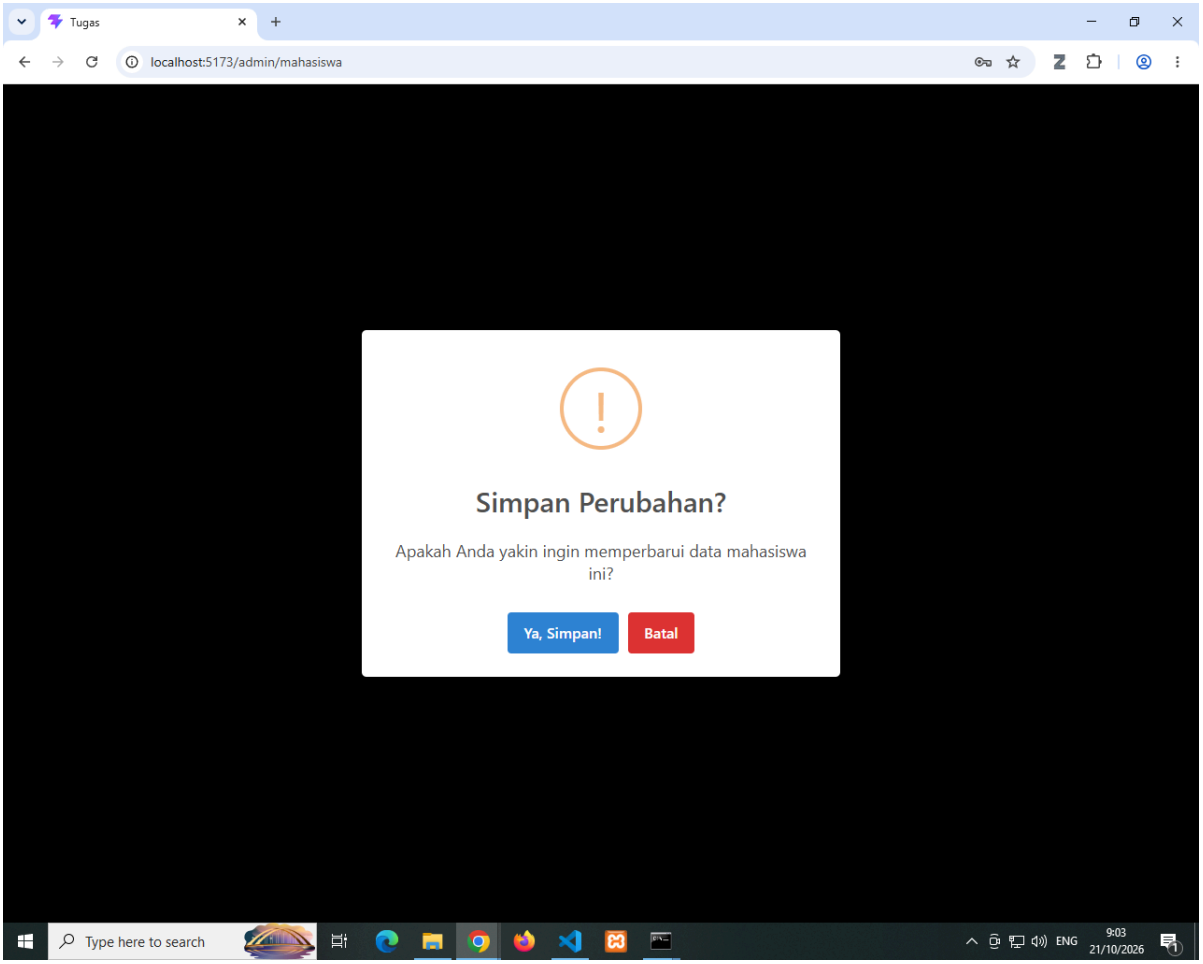
B. Pop-up Konfirmasi Logout (SweetAlert2)



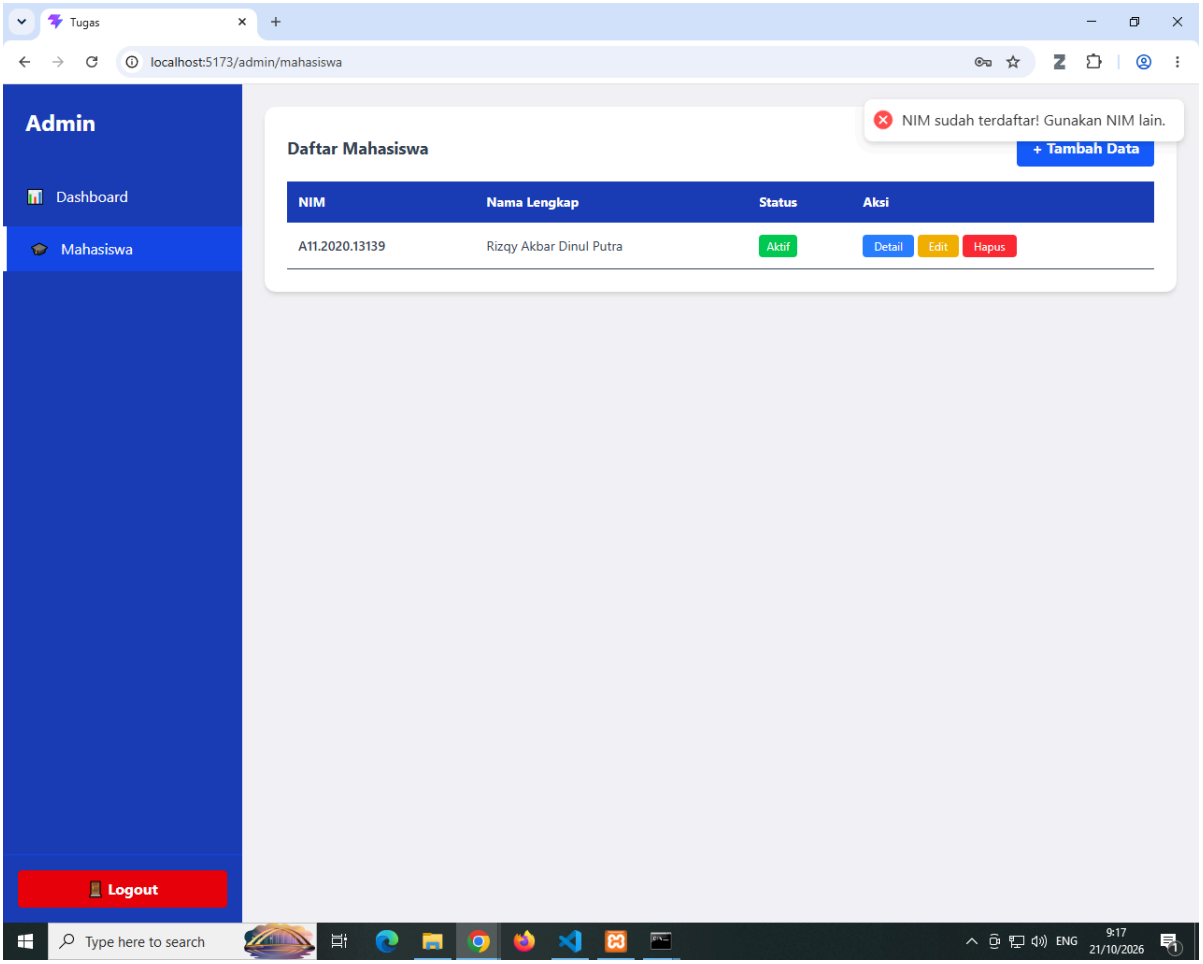
C. Pop-up Konfirmasi Hapus Data Mahasiswa (SweetAlert2)

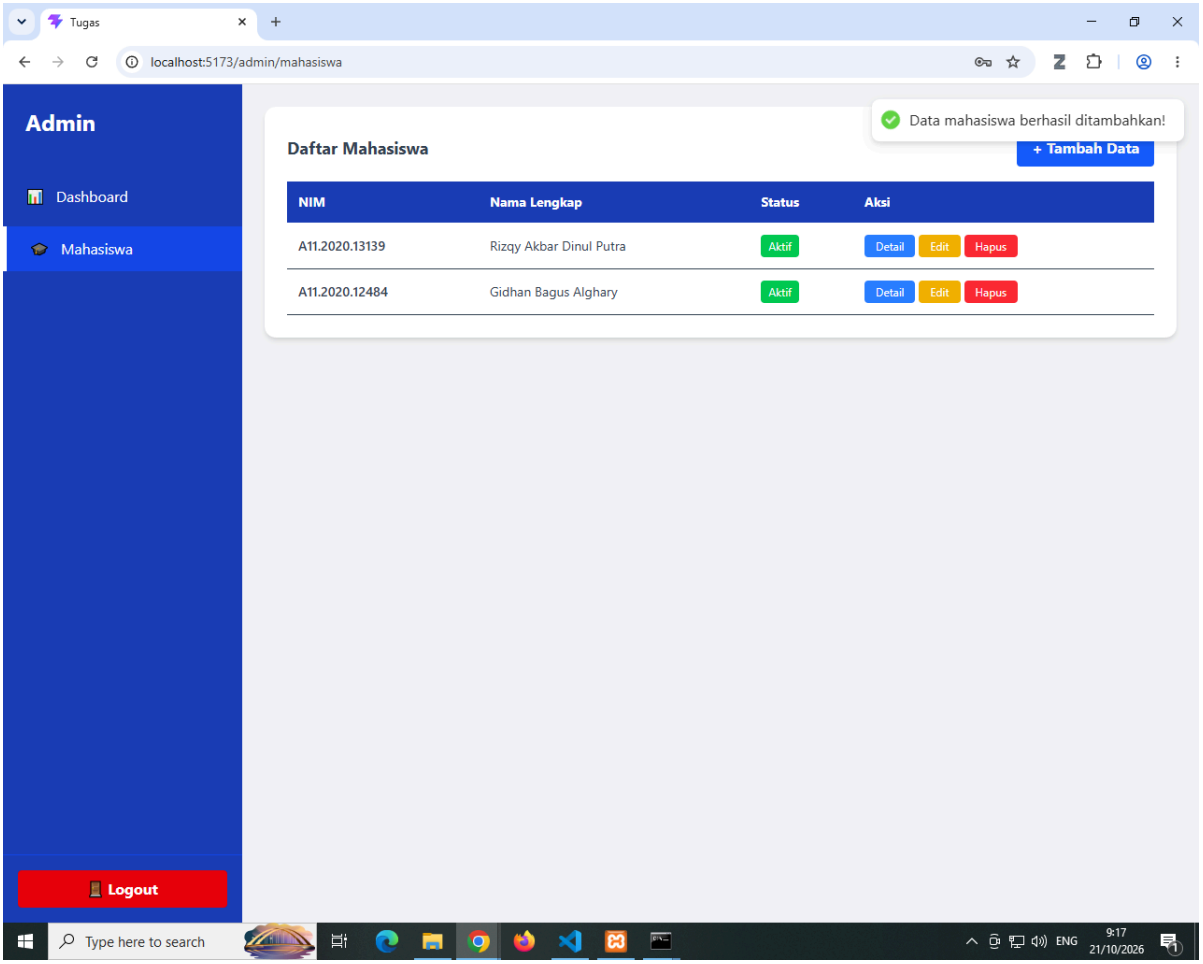


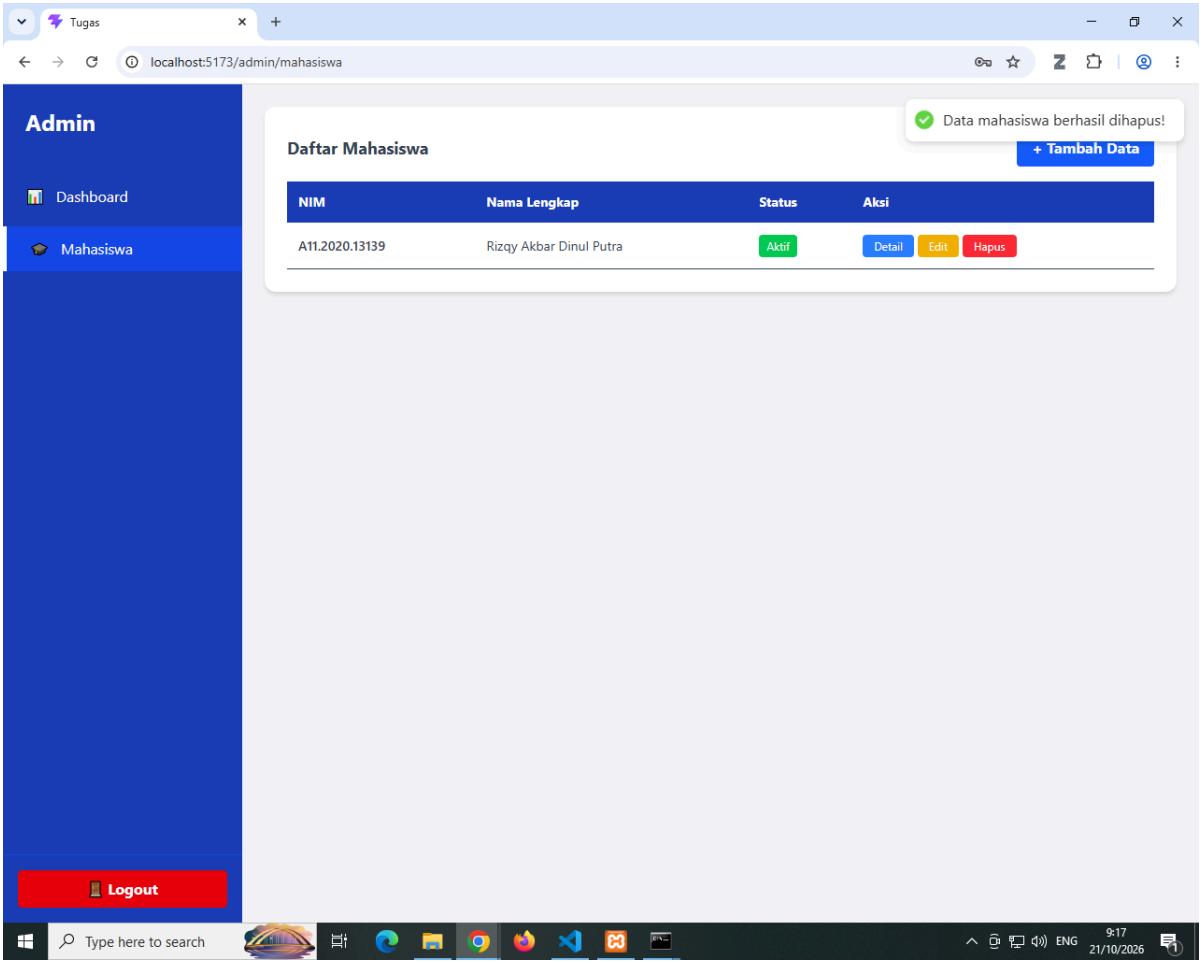
D. Pop-up Konfirmasi Simpan Perubahan atau Update Data (SweetAlert2)



E. Toast Feedback Tambah atau Update atau Hapus Mahasiswa


Tugas
localhost:5173/admin/mahasiswa
Admin
Dashboard
Mahasiswa
Logout
NIM sudah terdaftar! Gunakan NIM lain.
Daftar Mahasiswa
+ Tambah Data
NIM
Nama Lengkap
Status
Aksi
A11.2020.13139
Rizqy Akbar Dinul Putra
Aktif
Detail
Edit
Hapus
Type here to search
ENG
9:17
21/10/2026


Tugas
localhost:5173/admin/mahasiswa
Admin
Dashboard
Mahasiswa
Logout
Data mahasiswa berhasil ditambahkan!
Daftar Mahasiswa
+ Tambah Data
NIM
Nama Lengkap
Status
Aksi
A11.2020.13139
Rizqy Akbar Dinul Putra
Aktif
Detail
Edit
Hapus
A11.2020.12484
Gidhan Bagus Alghary
Aktif
Detail
Edit
Hapus
Type here to search
ENG
9:17
21/10/2026



4. Kesimpulan

Pada Tugas 7 ini, interaksi antarmuka pengguna pada aplikasi web React telah berhasil ditingkatkan dengan mengintegrasikan library SweetAlert2 dan React Hot Toast. Seluruh dialog konfirmasi sensitif (seperti penghapusan data, pembaruan data, dan logout) ditangani secara interaktif melalui modul helper SweetAlert2, sementara umpan balik aksi pengguna (seperti status login dan notifikasi CRUD) disajikan secara dinamis dan non-intrusif menggunakan toast helper yang terpasang secara global pada aplikasi.